TEAM_CONTEXT2 — handoff after the diagnostic phase

Amazon ML Challenge 2026 · Business Entity Resolution · Team camelcase. Compiled 2026-09-26 at the end of the working day.

- **Read this first.** It extends `TEAM_CONTEXT.pdf`, which covers the pipeline and everything up to the first submission, with every experiment run since.
- **Sources:** all numbers come from the completed reports listed in the file map. Nothing here was recomputed.

Evidence labels used throughout:

- A measured local validation, labelled
- B measured test-like or stress environment, labelled
- C unlabelled real-test observation
- D actual leaderboard score
- E hypothesis or estimate

An E number is never a measured leaderboard result.

Labelled environments used by the diagnostics (all use the 441,364 validation S1 and their labels):

- **V:** normal validation (validation pool, 4.68 pool records per S1).
- **T:** test-ratio pool (validation pool plus random fit-fold records, 0.174 S1 per pool record, the same as the real test). Its target-context profile matches the real test.
- **S:** full 10.3M training pool, test-sized but orphan-heavy (0.043 S1 per pool record). Blocking on the real test behaves like S.

"Step 3 model" means `experiments/matching/model_gbdt.npz`. The submitted **final model** (`experiments/final/model_final.npz`) was trained on all labels, so it cannot be scored on labelled data.

CURRENT BEST-KNOWN STATE

Item	Value
Local validation (official macro F0.5)	A **0.962847** (Step 3 GBDT, threshold 0.69, one-S1-per-target)
Leaderboard: production submission	D **≈ 0.924** (`output/matching_results.tsv`, threshold 0.69). The exact decimals were not recorded in project files.
Leaderboard: 0.72 probe	D **0.923607**, the same result as production within noise
Current production threshold	**0.69** (unchanged)
Final matcher	From-scratch numba histogram GBDT: 300 trees, depth 6, learning rate 0.1, λ 1, minimum child weight 20, 64 bins, seed 0. 50 features. Retrained on all training S1 × the full training pool.
Blocking recall, validation (V)	A 98.214% (27,277 true pairs missed)
Blocking recall, test-sized pool (S)	B 95.631% (66,750 missed)
Test-ratio environment T	B 0.95432 (current model)
Stress environment S	B 0.84002 (current model; orphan-driven false positives)
Main known sources of loss (from validation 0.963 to the leaderboard ≈ 0.924)	• Test ratio / orphan competition: B −0.0085 • Test-sized blocking: B −0.0063 • Country mix (India-heavier test): B −0.0038 • Lookalike singletons: E −0.002 to −0.009 • France (unlabelled): E unknown; it would need about 0.80–0.87 for the total to reach 0.924 • Residual: E 0 to about 0.015, unexplained
Production artifacts	**Unchanged** since the submission (hashes in the production-safety section)

PHASE 1 — Existing pipeline (details in TEAM_CONTEXT.pdf)

Step	What	Key result
Step 1: validation and scorer	Leakage-safe 80/20 split at S1 level (seed 20260925, stratified); official macro-F0.5 scorer; tests	A 441,364 validation S1; 5.585% singletons; validator PASS
Step 2: blocking	TF-IDF name+address top-10 and address top-10 per source, within the query's country, max_postings 20,000; 16 exact-key joins with block cap 100 (config SHA `91a38976…0e5f5`)	A recall 98.214%, 58.5 candidates per S1; B 95.631% on the full pool
Step 3: matcher	50 features (keys, name/address similarity, TF-IDF, candidate rank, target context `t_rank`/`t_gap`/`t_ncand`); GBDT chosen on A2	A 0.962847 (logistic regression 0.94458); threshold 0.69, one-S1-per-target
Final training, inference and submission	Retrained on all training data; 126.3M test candidates	C 5,503,470 predictions; 110,945 S1 (6.40%) with no match; validator PASS; zip built

PHASE 2 — Completed diagnostics (all read-only; production untouched)

1. `t_rank` diagnostic CLOSED

- **Question:** is `t_rank` (76.5% of GBDT gain) behind the gap?
- **Experiment:** five variants retrained on the internal split (diagnostic only; no production retrain or inference):
  - A: current model
  - B: without `t_rank`
  - B2: without all three target-context features
  - C: `t_rank` capped at 1
  - D: percentile `t_rank`

  Each was evaluated on V, T and S. The test-ratio environment T was introduced here.
- **Results:**
  - B V / T / S for A: 0.96285 / 0.95432 / 0.84002.
  - B, C and D stay within ±0.004 of A everywhere; `t_gap` takes over from `t_rank`.
  - B2: 0.95527 / 0.95125 / 0.90154.
  - The best threshold rises with orphan density (about 0.71 in V, 0.77 in T, 0.96 in S).
- **Conclusion:** `t_rank` alone is not the cause. The target-context family is orphan-sensitive but still net-positive at the test ratio. `experiments/diagnostics/trank/TRANK_DIAGNOSTIC_REPORT.md`

2. Leave-one-country-out (LOCO) CLOSED

- **Question:** how badly does the matcher degrade in an unseen country, as France is?
- **Experiment:** US-only, India-only and size-matched US-only models, trained with the Step 3 protocol (diagnostic retraining, internal set only), evaluated per country on V, T and S.
- **Results** A / B:
  - **India → US:** −0.011 on V and −0.015 on T. It over-merges: false positives double, singleton F0.5 drops about 10 points, and the best threshold moves up.
  - **US → India:** −0.078 on V and −0.076 on T. Recall collapses (79% vs 92%).
  - Per-country models gain at most about 0.001 in-country.
- **France, unlabelled** C:
  - Fewest S1 with no match (4.62% vs US 6.13%, India 7.20%).
  - Generic names are not left empty more often.
  - Very generic names (e.g. "Bordeaux Club SAS" ×516).
- **Conclusion:** unseen-country transfer is real but asymmetric. France's predictions show the over-merge pattern, which is suggestive only. `experiments/diagnostics/leave_one_country_out/LOCO_REPORT.md`

3. France-only threshold CLOSED

- **Question:** would a higher threshold for France only help?
- **Experiment:** filter the submitted predictions for France S1 at 0.80, 0.82, 0.85 and 0.88. No retraining or inference.
- **Results:**

- C 0.80 removes 23,683 France predictions (2.59%), and France S1 with no match go from 4.62% to 4.96%. 0.88 removes 48,481 (5.31%).
- A / B In the labelled analogues, 54–85% of removed predictions were true matches.
- E Overall effect about ±0.001.

- **Conclusion:** a weak lever; do not pursue. `experiments/diagnostics/france_threshold/FRANCE_THRESHOLD_REPORT.md`

4. Target-context / `t_rank` ablation CLOSED

- **Question:** does removing or neutralising the target-context features fix the shift?
- **Experiment:** reused the `t_rank` diagnostic's models; added entity metrics and feature distributions, including the real test. No retraining or inference.
- **Results:**
  - B Validation-to-stress gap: A 0.1228; without `t_rank` 0.1253; without all target context 0.0537. That comes at −0.0076 on V and −0.0031 on T.
  - C Real-test target-context distributions are close to V and T (PSI ≤ 0.072), far from S (0.56–1.17).
  - 99.8% of true positives have `t_rank` = 0: it works as a soft "best claimant" rule.
- **Conclusion:** keep the current matcher.
  `experiments/diagnostics/target_context_ablation/TARGET_CONTEXT_ABLATION_REPORT.md`

5. Blocking-gap investigation CLOSED

- **Question:** could candidate generation explain the gap?
- **Experiment:** analysis of saved candidate sets plus a hybrid estimate. No retraining or inference.
- **Results:**
  - C The real test's candidates per S1 (mean 72.9, p99 255) match S, not T. Blocking depends on absolute pool size.
  - B A perfect matcher on the candidates would score: V 0.99386, T 0.99305, S 0.98453.
  - Test-sized blocking costs −0.0063 (T 0.9543 → hybrid 0.9480).
  - Weakest categories at test size: non-Latin 80.8%, empty address 78.5%, house number differs 88.9%, generic names (India 88.1%).
- **Conclusion:** blocking is healthy but costs about 0.006; do not modify.
  `experiments/diagnostics/blocking_gap/BLOCKING_GAP_REPORT.md`

6. Deep France error analysis CLOSED

- **Question:** which France-like error patterns could explain the gap?
- **Experiment:** analogue profiling, France-profile reweighting, and a category-mix precision estimator validated on the LOCO cases. No retraining or inference.
- **Results:**
  - The closest analogue is US in T.
  - C France-specific traits:
    - 4.2% of France S1 share a name with 100+ others
    - initialism names ("SC") make up 1.74% of the pool, 9–17× the other countries
    - few house-number conflicts
  - The implied precision of France's category mix is 0.969, no worse than the US.
  - Fewer S1 with no match is a test-wide pattern.
- **Classification:** D, unresolvable, with weak signs of over-merging.
- **Conclusion:** no France-specific change is justified.
  `experiments/diagnostics/france_error_analysis/FRANCE_ERROR_ANALYSIS_REPORT.md`

7. Orphan-aware training CLOSED

- **Question:** does adding orphaned targets to training improve robustness at the test ratio?
- **Experiment:** internal set A plus 475,494 fit-fold orphans, bringing the pool to 5.754 per S1; retrained with the Step 3 protocol (diagnostic only, not production). The threshold chosen on A2 was 0.69.
- **Results** B:
  - V: 0.96172 (−0.0011). T: 0.95652 (+0.0022). S: 0.87194 (+0.032).
  - At matched V, the T gain is only +0.0008.
  - Target-context gain falls from 82.8% to 55.1%.
- **Conclusion:** most of the gain is just conservatism; not worth a submission. Stop this family.
  `experiments/diagnostics/orphan_aware_training/ORPHAN_AWARE_TRAINING_REPORT.md`

8. Train-vs-test distribution audit CLOSED

- **Question:** is the gap a population difference?
- **Experiment:** distribution tables, a test-matched labelled subset and singleton calibration. No retraining or inference.
- **Results:**
  - C S1 shares: US 60%/India 40% in train vs US 38.3%/India 46.8%/France 15.0% in test. Pool per S1 is +23–24% (test 5.53–5.82 vs 4.67–4.68). Source mix and missing addresses are not shifted.
  - C The largest new shift does not depend on the model: only 0.07–0.31% of test S1 lack a candidate with cosine ≥ 0.5, against 1.8–3.2% in V/T. Either there are fewer singletons, or singletons have lookalikes.
  - C Under the Step 3 model, test has +7% predictions per S1 and roughly double the [0.69, 0.80) band.
  - B The test-matched subset scores 0.9523 in T and 0.9459 with blocking losses.
  - B Lookalike singletons in T: F0.5 0.854 (US) / 0.793 (India), against 0.894 / 0.854 for all singletons. E A cost of −0.002 to −0.009.
- **Conclusion:** no single shift explains about 0.03. France would need about 0.80–0.87.
  `experiments/diagnostics/test_distribution_audit/TEST_DISTRIBUTION_AUDIT_REPORT.md`

9. Global threshold diagnostic CLOSED

- **Question:** should the global threshold be raised?
- **Experiment:** re-applied the rule to the saved final and Step 3 probabilities. No retraining or inference.

Threshold	Test predictions C	Removed	S1 with no prediction	V change A	T change B	S change B	Test-like scenario E
0.69	5,503,470	—	6.40%	—	—	—	—
0.70	5,483,773	0.36%	6.47%	+0.0001	+0.0004	+0.003	≤ +0.0006
0.72	5,444,249	1.08%	6.60%	+0.0000	+0.0009	+0.008	+0.0001 to +0.0014
0.75	5,382,517	2.20%	6.81%	−0.0003	+0.0014	+0.017	−0.0004 to +0.0021
0.78	5,317,344	3.38%	7.03%	−0.0010	+0.0016	+0.026	−0.0016 to +0.0022
0.80	5,271,154	4.22%	7.19%	−0.0017	+0.0014	+0.032	−0.0026 to +0.0020

- 46–53% of removed test predictions are house-number conflicts.
- **Conclusion:** keep 0.69; 0.72 is the only low-risk probe.
  `experiments/diagnostics/global_threshold/GLOBAL_THRESHOLD_REPORT.md`

PHASE 3 — Controlled leaderboard probe CLOSED

Item	Value
Probe	Global threshold **0.72**, re-applied to the saved final-model probabilities with the unchanged one-S1-per-target rule. No retraining or inference.
File	`experiments/leaderboard_probe_072/matching_results_072.tsv`: 92,615,809 bytes, SHA-256 `77559ba89f662cb59991a87eee7a7a05ecad2d14b7c4f8e0c50f7b49f9b87a4f`. Official validator PASS with `--check-ids`.
Change vs production	C • **59,221 predictions removed** (−1.08%), 0 added; 5,444,249 remain. • S1 with no match: 110,945 → 114,353. • US −17,107, India −35,896, France −6,218.
Leaderboard	D **0.923607**, against production ≈ 0.924. **No material change.**
Production vs probe	The production submission (`output/matching_results.tsv`, threshold 0.69) is **unchanged**. The probe is a separate experimental file; the user uploaded it manually for evaluation.
Conclusion	Threshold adjustment does not move the leaderboard. **All threshold work is stopped**: no 0.73, 0.75, 0.78, 0.80 or France thresholds.

PHASE 4 — Latest: blocking-miss analysis CLOSED (analysis)

- **Question:** are the true pairs lost by blocking a realistic opportunity?
- **Experiment:** features for missed pairs only, exact-channel cap checks, and targeted retrieval re-runs only for the S1 with misses plus a 20,000-S1 sample for cost. No retraining, no inference, no full regeneration.
- **Recall:** A V 98.214% (27,277 missed). B S 95.631% (66,750 missed; S2 32,972 / S3 33,778; US 32,366 / India 34,384).

Failure mode (S, mutually exclusive)	Misses	Share	Budget-200k channel recovers
Target name non-Latin (script differs)	21,278	31.9%	34 (address K=20 recovers 2,547)
House number differs	18,316	27.4%	8,923
Target address empty	14,310	21.4%	5,230
Other partial name/address differences	8,584	12.9%	4,693
Name shares no token (noisy/domain)	3,566	5.3%	16
Address dissimilar	696	1.0%	236

Retrieval variant (tested on the misses)	S recovered	V recovered	New candidates per S1 (S / V)
Name+address K=20 (K=50)	7,316 (15,162)	3,650 (7,306)	16.6 (69.9) / 17.0
Address K=20	4,687	3,355	17.5 / 17.0
New name-only K=10	1,519	172	10.8 / 7.5
Name+address top-10, max_postings 200,000	**19,132 (28.7%)**	4,626 (17.0%)	**6.7 / 5.8**

- The exact-key misses in S were all dropped by the 100-record cap (e.g. house+addr_digits 7,122). Raising the cap is not efficient.
- **Proposed, NOT implemented:** an **additional** name+address channel (top-10 per source, max_postings 200,000) alongside the existing one.
  - Add it rather than replace the existing budget: replacing swaps candidates, and the Step 2 design saw standalone recall fall at larger budgets.
  - E S recall 95.63% → about 96.88%.
  - Cost about +6.7 candidates per S1 (+9%; about +11.6M test pairs); retrieval about 6× slower per query.
  - E End-to-end gain about +0.001 to +0.003 F0.5.
- `experiments/diagnostics/blocking_miss_analysis/BLOCKING_MISS_ANALYSIS_REPORT.md`

CLOSED EXPERIMENTS — DO NOT REPEAT

Future sessions must **not** rerun or repeat any of these unless there is an explicit new reason:

1. Step 1 split and scorer, Step 2 blocking design, Step 3 matcher selection, final training and inference, submission and zip
2. Labelled stress evaluation (S), and the unlabelled test-scale check
3. `t_rank` diagnostic (variants B, B2, C, D; environments V/T/S)
4. Leave-one-country-out (US-only, India-only, size-matched)
5. France-only threshold (0.80–0.88)
6. Target-context / `t_rank` ablation
7. Blocking-gap investigation (including the hybrid estimate)
8. Deep France error analysis
9. Orphan-aware training (a whole family; do not propose variants)
10. Train-vs-test distribution audit (including the test-matched subset)
11. Global threshold diagnostic (0.70–0.80)
12. 0.72 leaderboard probe (scored 0.923607). **No further threshold probes.**
13. Blocking-miss analysis (as analysis; its proposed follow-up is open, below)

REMAINING TECHNICAL OPTIONS (unranked)

Option	Expected upside	Expected compute	Known risks	Evidence that would justify proceeding
1. Additional blocking channel (name+address top-10, 200,000 postings)	E About +0.001 to +0.003 F0.5; recovers about 29% of test-sized misses	Step 1: S-environment end-to-end test, about 30–60 min (new channel for 441k validation S1; features and target context on the enlarged set; score with the existing Step 3 model). A production rebuild would add about 2–3 h (test candidates, features, inference), plus ideally retraining.	Recovered pairs are hard (house-number and address differences), so matcher precision on them is lower. +6.7 candidates per S1 add false-positive chances. Slower retrieval.	An S end-to-end macro F0.5 of at least about +0.002 over 0.84002, and no loss in T
2. Alternative matcher using the existing candidates and features	E Unknown. Only GBDT tree count and logistic regression were ever compared; depth, learning rate, λ and bins were never tuned.	Training on the saved A1 features (`experiments/matching/X_A.npy`) takes minutes per model; scoring the saved V/T/S matrices is cheap	Overfitting to A2; small gains that sit below leaderboard resolution; the leaderboard gap appears to be mostly population- and France-driven	A gain on T of at least about +0.002 with no loss on V, chosen on A2, not on V or T

TOMORROW — START HERE

1. Read **TEAM_CONTEXT2.pdf** (this file). `TEAM_CONTEXT.pdf` is only needed for pipeline details.
2. **Do not rerun any closed experiment.**
3. Decide between exactly two options:
   - the proposed blocking channel (S end-to-end test first), or
   - a controlled alternative-matcher experiment on the existing features.
4. The API budget is limited. Run **one tightly scoped experiment at a time**, and **stop if it fails to beat its baseline** (T 0.95432 / V 0.962847 / S 0.84002).
5. Never overwrite `output/`, `experiments/final/`, `submission_package/` or the zip. Write new outputs to a new directory, and hash production before and after.

PRODUCTION SAFETY

Artifact	Latest recorded SHA-256 / status
`output/matching_results.tsv` (production submission, threshold 0.69)	`84ede6d4f2e44ab4469ea7798a95fbd8b3c49fc0f3f1561d47064c1caaa0e807`

`output/candidate_pairs.tsv`	`1842a3df05f69fd9236e4601cef27ded0d18e7e2c481dd803f27e9f8ff4835ea`
`experiments/final/model_final.npz`	`1bac911596c3d42eca3dc0c0c01660af3eb4bd6fe27f5e68299f3f4c02866755`
`experiments/final/` (168 files)	All hash-identical before and after every diagnostic (last checked by the blocking-miss analysis)
`camelcase_submission.zip`	`93b8db37fd2c8a79d74cc80ceb8051451b546f817393b5b1278f76426ed374a8`
`submission_package/`	Unchanged; included in the 34 protected files checked by each diagnostic
Blocking config	`91a3897685414bb48c4bc4f00a162e1b245eb8640166f950a76bdc379042e0f5`

- No production artifact was modified by any diagnostic.
- The 0.72 probe is a separate experimental file.
- No new blocking has been implemented or applied.
- No new final submission has been built since the 0.72 probe.

FILE MAP

What	Where
Handoff documents	`TEAM_CONTEXT.pdf`, `TEAM_CONTEXT2.pdf` (project root); sources `docs/TEAM_CONTEXT.html`, `docs/TEAM_CONTEXT2.html`
Rules and data facts	`docs/COMPETITION_NOTES.md`, `docs/problem_statement.pdf`
Pipeline reports	• `experiments/validation_split/VALIDATION_SPLIT.md` • `experiments/blocking/BLOCKING_REPORT.md` • `experiments/matching/MATCHING_REPORT.md` • `experiments/TEST_SCALE_REPORT.md` (stress) • `experiments/test_scale/TEST_SCALE_REPORT.md` (unlabelled test) • `experiments/final/FINAL_REPORT.md`
`t_rank` diagnostic (V/T/S models, probabilities, feature matrices `X_S`/`X_T`)	`experiments/diagnostics/trank/`; script `experiments/trank_diagnostic.py`
LOCO	`experiments/diagnostics/leave_one_country_out/LOCO_REPORT.md`; script `experiments/loco_diagnostic.py`
France threshold	`experiments/diagnostics/france_threshold/FRANCE_THRESHOLD_REPORT.md`
Target-context ablation	`experiments/diagnostics/target_context_ablation/TARGET_CONTEXT_ABLATION_REPORT.md`
Blocking gap	`experiments/diagnostics/blocking_gap/BLOCKING_GAP_REPORT.md`
France error analysis	`experiments/diagnostics/france_error_analysis/FRANCE_ERROR_ANALYSIS_REPORT.md`
Orphan-aware training	`experiments/diagnostics/orphan_aware_training/ORPHAN_AWARE_TRAINING_REPORT.md`
Distribution audit	`experiments/diagnostics/test_distribution_audit/TEST_DISTRIBUTION_AUDIT_REPORT.md`
Global threshold	`experiments/diagnostics/global_threshold/GLOBAL_THRESHOLD_REPORT.md`
0.72 probe	`experiments/leaderboard_probe_072/` (`matching_results_072.tsv`, `PROBE_REPORT.md`)
Latest blocking-miss analysis	`experiments/diagnostics/blocking_miss_analysis/BLOCKING_MISS_ANALYSIS_REPORT.md` (and `results.json`, `blocking_miss_analysis.py`)
Production submission	`output/matching_results.tsv`, `output/candidate_pairs.tsv`, `camelcase_submission.zip`, `submission_package/`, `experiments/final/`
Smoke-scale copies for quick code tests	`experiments/_smoke/`

Each diagnostic directory also contains its script, `results.json`/`evaluate.json`, `run.log` and hash files.